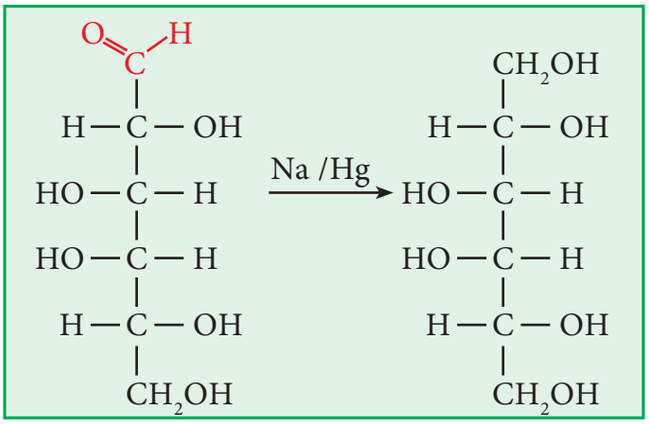
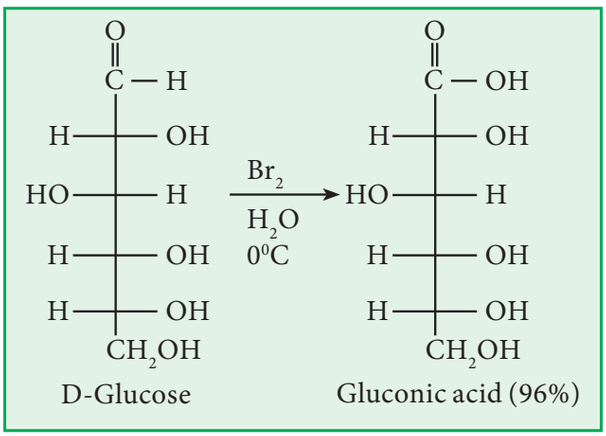


v. Oxidation:

a. Glucose is oxidized by mild oxidizing agents like bromine water into gluconic acid. Strong oxidizing agents like conc. HNO_3 oxidize glucose into gluconic acid.



b. Fructose is not oxidised by mild oxidising agents. But strong oxidising agents like conc. HNO_3 split fructose into a mixture of trihydroxy glutaric, tartaric and glycollic acids.

c. Galactose on oxidation with mild oxidizing agent such as bromine water, gives galactonic acid. On oxidation with strong oxidizing agents like HNO_3, it gives galactaric (or) mucic acid. This acid is insoluble in water and hence this reaction is used as a test for galactose. On oxidation with O_2/Pt-C (as in glucose, after protecting the –CHO group into ispropylidene group) it gives galacturonic acid.

vi. Action with Tollen's reagent:

Both glucose and fructose reduce Tollen's reagent into silver mirror.

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